



Figure 1: Methodology of RNA Seq Analysis: Pipeline outlines the entire process to obtain the significant genes. Original figure made with PowerPoint.

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25686941 reads; of these:
25686941 (100.00%) were paired; of these:
  1645664 (6.41%) aligned concordantly 0 times
  22451232 (87.40%) aligned concordantly exactly 1 time
  1590045 (6.19%) aligned concordantly >1 times
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  1645664 pairs aligned concordantly 0 times; of these:
    251855 (15.30%) aligned discordantly 1 time
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  1393809 pairs aligned 0 times concordantly or discordantly; of these:
    2787618 mates make up the pairs; of these:
      1497905 (53.73%) aligned 0 times
      1139371 (40.87%) aligned exactly 1 time
      150342 (5.39%) aligned >1 times
97.08% overall alignment rate
  
```

Figure 2: An example output of HISAT2. It provides the reads which were paired, aligned and unpaired along with the percentage of overall alignment. Image from galaxy.eu.

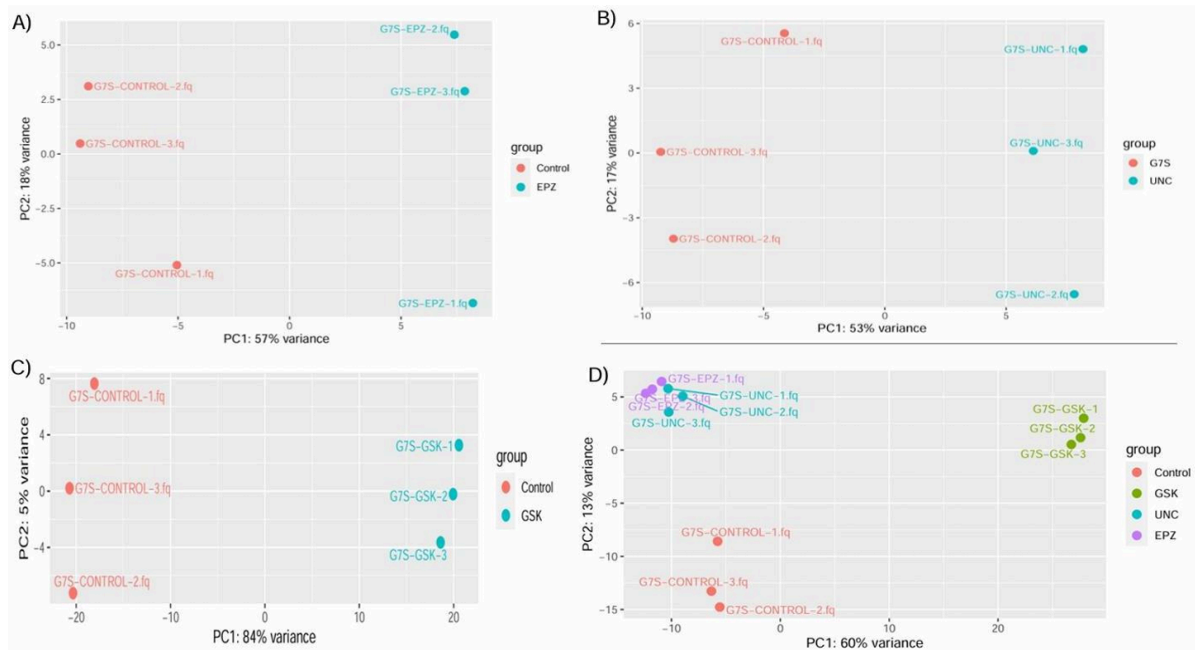


Figure 3: The PCA biplots were obtained with controls plotted against the three drugs. (A) Control vs EPZ6438: Among the replicates of EPZ 6438, EPZ 2 and 3 were aligned with Control 2 and 3 in the same direction whereas, EPZ 1 and Control 1 were aligned in the same direction. When components were grouped in circles, they don't overlap each other. **(B) Control vs UNC1999:** Among the replicates of UNC 1999 were aligned in. UNC 1 and 3 were aligned with Control 1 and 3 in the same direction whereas, UNC 2 and Control 2 were aligned in the same direction. When components are grouped in circles, they don't overlap each other. **(C) Control Vs GSK343:** Among the replicates of GSK343, GSK 1 and 2 were aligned with Control 1 and 3 in the same direction whereas, GSK 3 and Control 2 were aligned in the same direction. When components are grouped in circles, they don't overlap each other. **(D) Control Vs Drugs (EPZ6438 vs UNC1999 vs GSK343):** From the PC1 axis, the GSK 343 triplicates were aligned in one direction whereas the control, EPZ 6438 and UNC 1999 triplicates were aligned on the other. From the PC2 axis, the triplicates of the drugs EPZ 6438, UNC 1999 and GSK 343 were aligned with in the same direction whereas, triplicates of the Control were aligned in the other direction. When components are grouped in circles control and GSK 343 didn't overlap each other whereas EPZ 6438 and UNC 1999 were closely related.

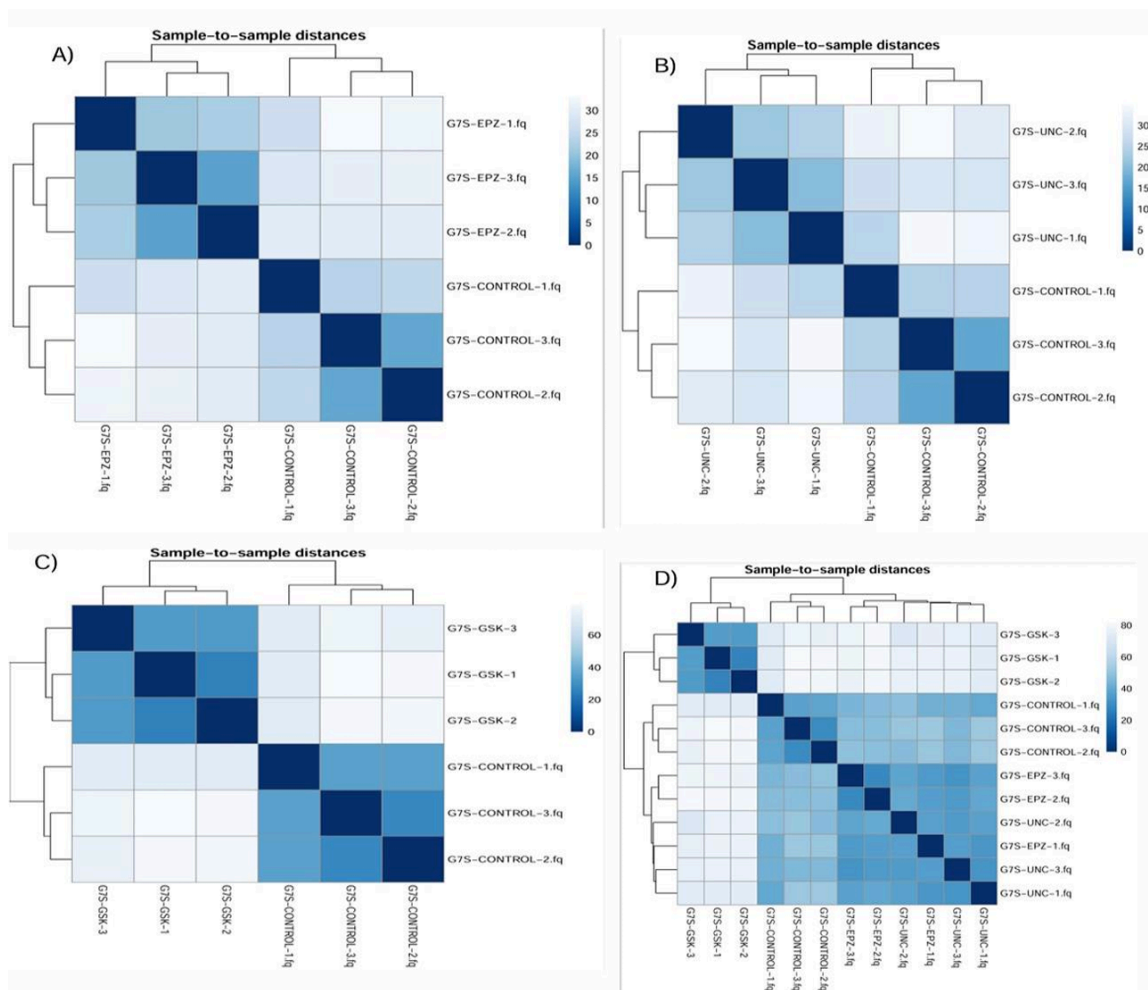


Figure 4: Heatmap of the Drugs Vs Control: (A) **Control vs EPZ6438:** The EPZ 6438 triplicates were correlated with itself and not highly with the triplicates of the control. (B) **Control vs UNC 1999:** The UNC 1999 triplicates were correlated with itself and not highly with the triplicates of the control. (C) **Control vs GSK 343:** The GSK 343 triplicates were correlated with itself and not highly with the triplicates of the control. (D) **Control Vs Drugs (EPZ 6438 vs UNC 1999 vs GSK343):** The triplicates of the drugs EPZ 6438 and UNC 1999 were correlated, whereas GSK 343 was found to be least related and all three were dissimilar from the control.

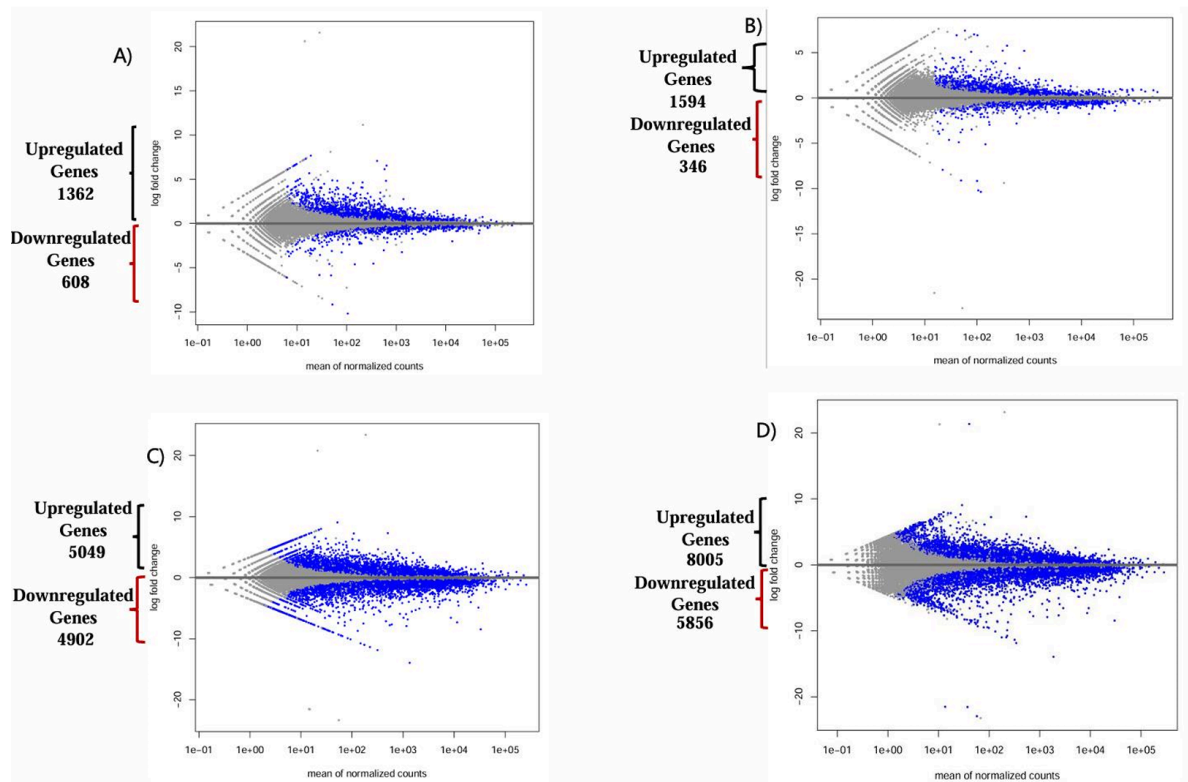


Figure 5: MA Plot : (A) **Control vs EPZ 6438:** The number of upregulated genes were 1362 and downregulated were 608. (B) **Control vs UNC 1999:** The number of upregulated genes were 1594 and downregulated were 346. (C) **Control vs GSK343:** The number of upregulated genes were 5049 and downregulated were 4902. (D) **Control vs Drugs (EPZ 6438 vs UNC1999 vs GSK343):** The number of upregulated genes were 8005 and downregulated were 5856.